# Enhancing Data and Algorithm in Offline Reinforcement Learning

**Tonghe Zhang**
Department of Electronic Engineering
Tsinghua University
Beijing, China
zhang-th21@mails.tsinghua.edu.cn

**Yucheng Pan**
Department of Physics
Tsinghua University
Beijing, China
panyuche21@mails.tsinghua.edu.cn

**Wei Zhang**
Department of Computer Science and Technology,
Tsinghua University,
Beijing, China
zhangw23@mails.tsinghua.edu.cn

**Yulong Xia**
Department of Computer Science and Technology,
Tsinghua University,
Beijing, China
xiayl23@mails.tsinghua.edu.cn

## Abstract

This work explores how the scale and structure of the dataset affect various algorithms in offline reinforcement learning. The primary goal is to deepen the understanding of established methodologies and cutting-edge solutions that enhance the performance of offline RL algorithms using limited data samples, as well as identify the key factors that affect the generalizability and robustness of offline RL algorithms. We conduct extensive experiments on three classical offline-RL algorithms and one state-of-the-art algorithm EDAC, investigating the effect of trimming the percentage of the dataset on the return of all the algorithms and discovering how different merging strategies affect the transfer learning ability of the agent when mixing original data with synthetic data created by diffusion models. We also try to explain how synthetic data influences the agent's exploitation and exploration.

## 1 Introduction

Reinforcement Learning (RL) has emerged as a powerful paradigm in machine learning, where an agent learns to make decisions by interacting with an environment to maximize cumulative reward. Traditional RL methods typically operate in an online setting, where the agent continuously interacts with the environment to gather data and improve its policy. However, many real-world applications such as autonomous driving, healthcare, and finance demand a more conservative approach due to safety concerns, high costs, or the infeasibility of constant interaction. This has given rise to the study and application of Offline Reinforcement Learning (Offline RL).

Offline RL aims to learn effective policies from a fixed dataset of previously collected experiences, without further interaction with the environment. This shift from online to offline learning brings

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several advantages, including enhanced safety and cost-efficiency, as it eliminates the need for real-time data collection. However, it also introduces significant challenges that necessitate specialized algorithms and approaches.

At the core of RL lies the concept of a Markov Decision Process, defined by a set of states, actions, state transition probabilities, reward function, and discount factor. An agent in an MDP aims to learn a policy that maximizes the expected cumulative reward. In the offline setting, the agent's learning is constrained to a fixed dataset composed of state-action-reward-next-state tuples collected from prior interactions, often through a suboptimal behavioral policy. The key challenge is to learn a robust policy from this static dataset without exacerbating distributional shifts and other issues inherent to offline data.

Several algorithms have been proposed to tackle the unique challenges of Offline RL. Batch Constrained Q-learning (BCQ) modifies the Q-learning framework to constrain actions to those seen in the offline dataset, mitigating the problem of extrapolation errors. Conservative Q-Learning (CQL) aims to learn conservative value estimates that are less likely to overestimate the value of out-of-distribution actions, thereby improving policy stability and performance. Implicit Q-Learning (IQL) learns a policy by implicitly estimating Q-values without explicit constraints on the policy. SAC-N, an extension of Soft Actor-Critic (SAC), introduces a normalization technique to handle the challenges posed by offline data.

Despite the progress, several key problems persist in Offline RL. Offline datasets often represent a limited and potentially biased sample of the environment's state-action space, leading to a distributional shift. Estimating the value of actions not present in the dataset can lead to significant extrapolation errors, as the learned Q-values may not generalize well outside the distribution of the training data. Moreover, offline datasets can exhibit reward bias, where certain state-action pairs are overrepresented or underrepresented, skewing the learning process.

This work is dedicated to exploring how the scale and structure of the dataset affect various algorithms in offline reinforcement learning. The primary goal of our research is to deepen the understanding of the established methodologies and cutting-edge solutions that enhance the performance of offline RL algorithms using limited data samples, as well as identify the key factors that affect the generalizability and robustness of offline RL algorithms.

Our contributions are as follows:

- We conducted extensive experiments on three classical offline-RL algorithms and one state-of-the-art algorithm EDAC, compared and analyzed their differences in the convergence rate and sample complexity on both the original and synthetic offline dataset.
- We analyzed the reasons behind the effect of augmenting the dataset with diffusion model-generated trajectories. In particular, we tried to explain how this method improves both the exploitation and exploration of the agent.
- We thoroughly investigated the effect of trimming percentage on all the algorithms, and discovered that we can compress the dataset without hurting the agent's generalizability when the trimming ratio is below a certain threshold.
- We discovered how different merging strategies can affect the transfer learning ability of the agent when we mix the original data with synthetic data created by the diffusion model. We also compared and analyzed how the four algorithms adapt to purely synthetic data, and how the ratio of fake data influences the average reward in an interesting fashion.

## 2 Related Work

**Trimming offline dataset.** Recent studies have introduced innovative dataset-trimming strategies to improve training efficiency and performance. DiffStitch [7] employs diffusion-based trajectory stitching to boost offline RL by enhancing data diversity and quality. Exploration by RND (Random Network Distillation) [3] enhances exploration capabilities by using a neural network to predict randomness, while the Survival Instinct strategy [6] focuses on the insensitivity of offline RL to manipulated reward signals, allowing more flexible reward engineering.

**Augmenting offline dataset.** The generation of synthetic trajectories using diffusion models has become a pivotal area of research. Synthetic Experience Replay (SER) [8] generates synthetic experiences unconditionally, demonstrating improved performance through periodic dataset updates, particularly in small datasets. Generative Trajectory Augmentation (GTA) [5] enhances data quality by generating high-reward trajectories and is effective in both dense and sparse reward settings. Policy-Guided Diffusion (PGD) [4] addresses the distribution shift between target and behavior policies by generating trajectories that are representative of the target policy while maintaining proximity to the behavior policy.

Building upon these advancements, our approach seeks to address the limitations identified in prior works. According to "Survival Instinct" [6], offline RL is not sensitive to reward signal manipulation, which can be leveraged to generate high-quality trajectories. Diffusion models, as demonstrated by DiffStitch [7] and many other works, can be used to generate data of high quality, approximate the real distribution and therefore significantly enhance generalization in offline RL. Our method extends the principles of GTA [5] by augmenting datasets with high-return trajectories while maintaining fidelity through a balanced mixture of synthetic and original data. These innovations aim to strike a balance between the diversity and fidelity of the data, thereby enhancing the overall performance of offline RL algorithms.

## 3 Methodology

**Baselines.** We use BCQ, CQL, and IQL as algorithms for evaluation. Batch-constrained deep Q-learning tries to prevent inquiring OOD actions by requiring a' to lie within the data distribution. In practice, this is implemented as an imitator policy by training a Conditional Variational Autoencoder(CVAE) on the behavior dataset. A learned perturbation is added to the samples of CVAE to do maximization of Q. Instead of constraining the policy, CQL conservatively estimates the Q value by pushing down the large Q values, especially on OOD state-action and performing standard critic updates. This avoids the potentially inferior modeling of the behavior policy as in BCQ. IQL estimates advantages using in-sample data only using executive regression of V and extracts the policy via advantage-weighted regression, completely avoiding any OOD query.

**SAC-n (EDAC).** The Ensemble Diversified Actor-Critic (EDAC) algorithm [1], addresses the challenges inherent in offline reinforcement learning (RL) where the training data is fixed and cannot be sampled further. The core idea of EDAC is to leverage an ensemble of Q-functions to estimate the uncertainty of the learned policy. By doing so, it prevents the learned policy from exploiting errors in Q-value estimation which commonly occur due to the limited and potentially biased offline data. EDAC utilizes a diversity-promoting mechanism within the ensemble to ensure the Q-functions offer varied perspectives on the value estimates. This mechanism penalizes correlations among ensemble members, allowing the algorithm to better detect and manage uncertainty. The approach balances between improving the policy's performance on genuine high-reward actions and avoiding overestimation that can arise from the bias in the offline dataset. By incorporating uncertainty estimation and promoting diversity within the Q-ensemble, EDAC achieves a robust performance on a wide range of offline RL tasks, demonstrating improved reliability and efficiency compared to existing methods. The pseudo-code of EDAC algorithm is presented in Algorithm 1.

Training EDAC on the original dataset, we obtain the loss curve and returns in Figure 1.

**Data augmentation in offline RL.** There has been an emergence of work utilizing the powerful generative modeling ability of diffusion models for data augmentation in offline RL settings. As a preliminary evaluation of the effectiveness of these data augmentation methods, we focus on SER and GTA. Below we provide a brief description of these two methods. SER models the distribution of single-step transition using a diffusion model, which is conceptually simple. GTA uses a more sophisticated process aiming at generating a high-reward trajectory. They use diffusion to model the distribution of (sub)trajectories conditioned on discounted return. The sampling process has special designs for offline RL. Unlike the typical diffusion model that starts denoising from Gaussian, they first add noise to the original trajectories, resembling an exploration behavior. The denoising starts from this partially noised state by conditioning on an amplified discounted reward of the original trajectory with classifier-free guidance to push the trajectory towards a potentially better region with a higher reward, resembling an exploitation behavior.

**Algorithm 1** Ensemble-Diversified Actor-Critic (EDAC)

1: Initialize policy parameters $\theta$, Q-function parameters $\{\phi_j\}_{j=1}^N$, target Q-function parameters $\{\phi'_j\}_{j=1}^N$, and offline data replay buffer $\mathcal{D}$

2: Sample a mini-batch $B = \{(\mathbf{s}, \mathbf{a}, r, \mathbf{s}')\}$ from $\mathcal{D}$

3: Compute target Q-values (shared by all Q-functions):

$$y(r, \mathbf{s}') = r + \gamma \left( \min_{j=1,\dots,N} Q_{\phi'_j}(\mathbf{s}', \mathbf{a}') - \beta \log \pi_\theta(\mathbf{a}' \mid \mathbf{s}') \right), \quad \mathbf{a}' \sim \pi_\theta(\cdot \mid \mathbf{s}')$$

4: Update each Q-function $Q_{\phi_i}$ with gradient descent using

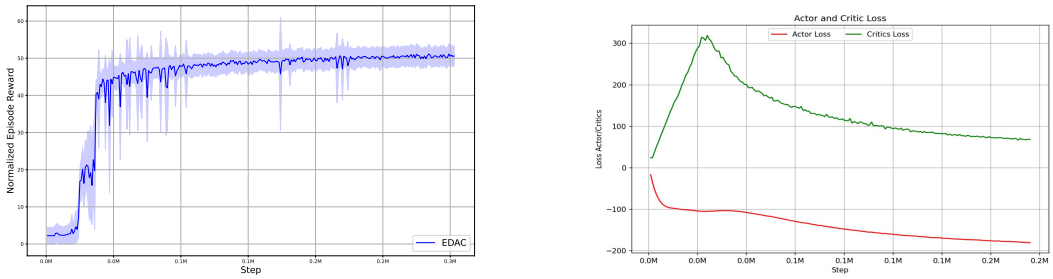
$$\nabla_{\phi_i} \frac{1}{|B|} \sum_{(\mathbf{s},\mathbf{a},r,\mathbf{s}') \in B} \left( \left( Q_{\phi_i}(\mathbf{s}, \mathbf{a}) - y(r, \mathbf{s}') \right)^2 + \frac{\eta}{N-1} \sum_{1 \le i \ne j \le N} \mathrm{ES}_{\phi_i, \phi_j}(\mathbf{a}, \mathbf{a}) \right)$$

5: Update policy with gradient ascent using

$$\nabla_\theta \frac{1}{|B|} \sum_{\mathbf{s} \in B} \left( \min_{j=1,\dots,N} Q_{\phi_j}(\mathbf{s}, \tilde{\mathbf{a}}_\theta(\mathbf{s})) - \beta \log \pi_\theta(\tilde{\mathbf{a}}_\theta(\mathbf{s}) \mid \mathbf{s}) \right),$$

where $\tilde{\mathbf{a}}_\theta(\mathbf{s})$ is a sample from $\pi_\theta(\cdot \mid \mathbf{s})$ which is differentiable w.r.t. $\theta$ via the reparametrization trick.

6: Update target networks with $\phi'_i \leftarrow \rho \phi'_i + (1 - \rho)\phi_i$




(a) Return (b) Loss

Figure 1: Results of training EDAC algorithm on the original offline dataset.

# 4 Experiments

## 4.1 Classical Algorithms

We use halfcheetah-medium-replay-v0 as the dataset. The dataset is composed of the replay buffer of a policy trained up to the performance of a medium agent with roughly 1/3 of a fully trained expert SAC agent. The dataset always timeouts at 1000 time steps. The dataset contains 101 trajectories. In total 101k transitions. The behavior policy is a mixture of random policy and medium policy. Therefore, the quality of trajectories varies much.

### 4.1.1 Trimming the Dataset for Classical Algorithms

We conduct experiments to measure and compare the performance of different models at trimming percentages from 0 to 0.5. Figure 2 shows the results. The normalized episode reward is computed as the actual episode return divided by the episode return of an expert policy. The scores are chosen to be the best environment evaluation score during training. We can see that the effect of dataset trimming varies for different algorithms. For BCQ, trimming leads to the greatest improvement. Hypothetically, the limited modeling ability of the CVAE imitator of BCQ fails to capture the behavior policy distribution for the dataset of mixed quality. For CQL, trimming leads to inferior performance

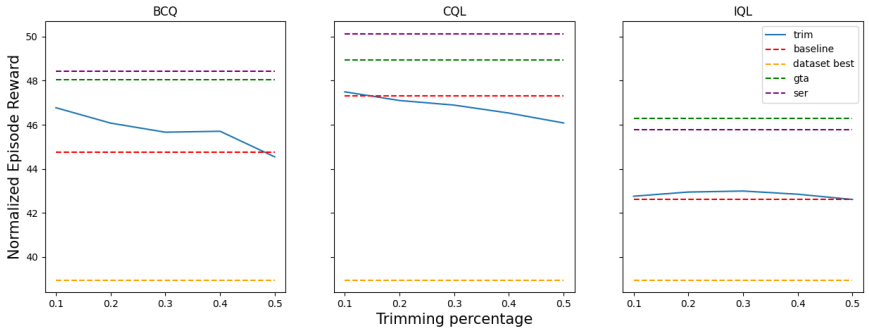



Figure 2: Results of trimming and augmentation on the dataset, training with BCQ, CQL and IQL.

in general. For IQL, the effect of trimming remains little, suggesting that IQL may be robust to the size and quality of the dataset.

#### 4.1.2 Data Augmentation with Diffusion Models for Classical Algorithms

We evaluate the performance of using SER and GTA for data augmentation. The configuration of data generation is in accordance with the official implementation. We generate 5M transitions using SER and GTA respectively. As the size of the dataset increases, the training for offline RL algorithms changes, as is done in SER and GTA. Specifically, the training steps are increased from 500K steps to 1M steps. For BCQ, it is observed that using the default KL loss coefficient for the CVAE imitator, as in the implementation of d3rlpy, leads to a consistent collapse. The imitator loss would be significantly larger than that of normal cases, indicating that the imitator may fail to model the behavior data when the size of the dataset is significantly increased. After experimenting, the coefficient is set to be 0.05. Interestingly, for CQL we also observed a sensitivity to hyperparameter setting in that decreasing the weight of the conservative loss could lead to a performance increase of about 5%. It's better to keep the hyperparameter setting unchanged anyway, as the absolute score is of little importance here.

We note that the performance of GTA is slightly inferior to that of SER, which contradicts the experimental results from GTA. The potential cause might be attributed to the difference in the dataset, as the evaluation in GTA is conducted on the v2 version of D4RL.

As a preliminary attempt at a better way to combine synthetic data with real data, we tried to adapt COMBO's loss for the model rollout to our synthetic data and mix the synthetic data with the real data as in COMBO. However, no significant improvement over conventional CQL is observed. It remains to see if this is a valid approach.

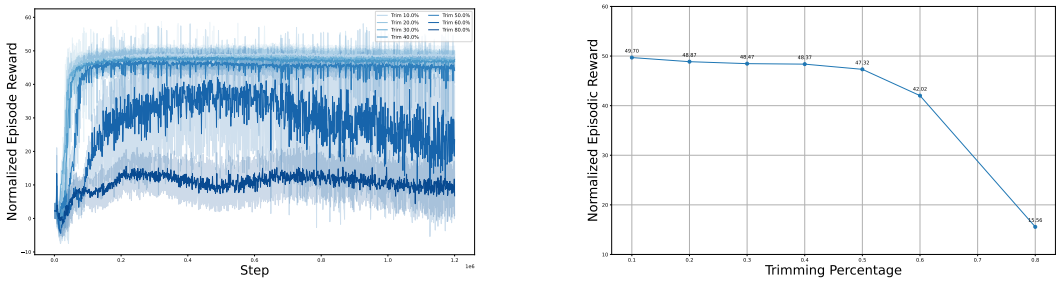
### 4.2 EDAC Algorithm

In what follows, we present experimental findings related to the use of the Efficient Diversified Actor-Critic (EDAC) algorithm, an advanced offline reinforcement learning (RL) method. EDAC is an efficient variant of the SAC-n (Soft Actor-Critic with n critics) algorithm. Our study focuses on evaluating the performance of EDAC under different experimental setups, particularly on large synthetic datasets, comparing it against other prominent offline RL algorithms such as BCQ (Batch-Constrained deep Q-Learning), CQL (Conservative Q-Learning), and IQL (Implicit Q-Learning).

One of the critical components of EDAC is its use of multiple critic networks (N critic nets), designed to address the issue of over-estimation in Q-learning. Additionally, EDAC includes a mechanism to maximize ensemble diversity, minimizing cosine similarity among the critic networks. This design ensures that the multiple Q-networks provide varied perspectives on the value estimates, thereby improving the robustness of the policy updates.

#### 4.2.1 Trimming the Dataset for EDAC Algorithm

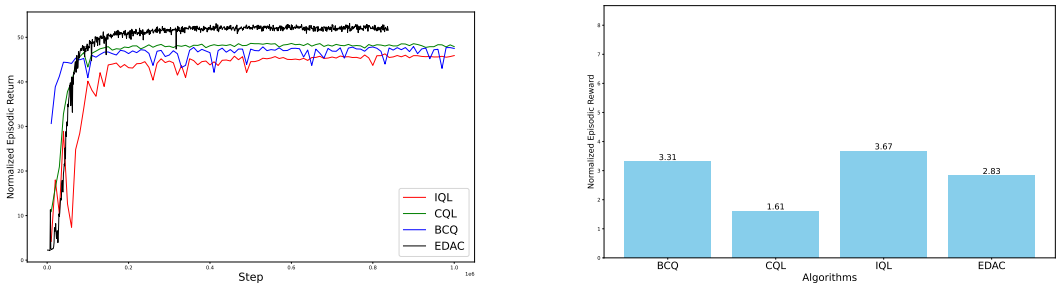
We sorted the episodes in the dataset according to their rewards and trimmed off the lowest rewarding episodes until the desired trimming percentage was reached. For example, a 0.1 trimming means




(a) Training NER curve.

(b) NER at different trimming percentages.

Figure 3: The effect of different dataset trimming percentages with EDAC algorithm.




(a) Returns on large synthetic dataset.

(b) Improvement of returns owing to synthetic data.

Figure 4: Different algorithm's performance on a synthetic dataset.

that we dropped $10\%$ lowest rewarding episodes. We tested different dataset trimming percentages with the advanced EDAC algorithm, and the results are shown in Figure 3. Note that the original EDAC performance without any dataset trimming is a NER of $50.3$. From Figure 3b we know that the performance of EDAC monotonically decreased when the dataset was trimmed to a smaller size, and the NERs were always lower than the NER of the untrimmed dataset. This could be due to the need for more data to effectively train the $N$ critic nets. However, we still found that trimming with a small percentage (less than $0.4$) will not drastically decrease the performance. It means that we need only $60\%$ of the transitions to achieve similar performance to the full dataset. Therefore this is a possible way to compress the storage of offline datasets without hurting performance too much.

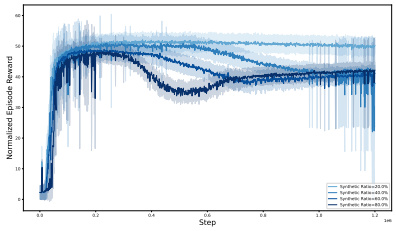
#### 4.2.2 Data Augmentation with Diffusion Models for EDAC Algorithm

In a series of experiments conducted on large synthetic datasets, EDAC demonstrated superior performance, outperforming BCQ, CQL, and IQL by a significant margin (see Figure 4). While BCQ and CQL showed similar spirits in handling offline data, IQL lagged considerably behind EDAC.

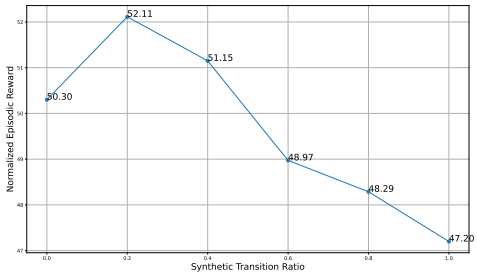
When comparing convergence rates, EDAC proved to converge much faster than the other algorithms on large synthetic datasets, with IQL being the slowest to achieve convergence. EDAC is highly data-driven in its current setup, requiring large amounts of data to operate efficiently. This is partly due to the algorithm's configuration with 10 Q-network heads, which make it parameter-intensive and consequently data-hungry. On smaller datasets, EDAC showed a tendency to overfit quickly, highlighting the need for abundant and diverse data.

Our initial experiments also revealed limitations when directly transitioning from synthetic to real data. We first trained the RL agent on a synthetic dataset, then directly moved it to learn the real environment. This method without shuffling led to suboptimal results (see Figure 5). Preliminary findings suggested that an abrupt distribution shift might be detrimental to the training process. This realization prompted a reconsideration of the experimental design.

To address these issues, we restructured our experiments by fixing the total number of transitions and adopting a more systematic data-handling approach. The results are shown in Figure 6. We began by selecting entire episodes from the diffusion dataset, generated within the same batch to maintain

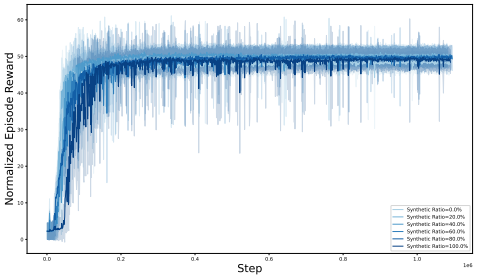



(a) Training NER curve.

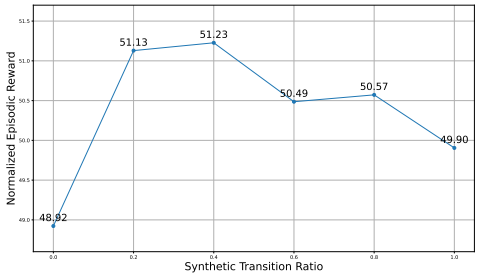



(b) NER at different synthetic ratio.

Figure 5: NER with direct transfer from synthetic transitions to real transitions at different mixture ratio of synthetic dataset.




(a) Training NER curve.




(b) NER at different synthetic ratio.

Figure 6: NER with smooth transfer from synthetic transitions to real transitions at different mixture ratio of synthetic dataset.

consistency. These episodes were then split into individual transitions. By randomly shuffling short (30, synthetic) and long (1000, real) transitions, we aimed to create a balanced mixture, thereby mitigating the risk of abrupt distribution shifts.

## 5 Analyses and Conclusions

### 5.1 Baseline Results

Data augmentation shows a strong ability to boost the performance of offline RL algorithms. This might feel strange as using a diffusion model to synthesize new data brings no additional "real" information. Here we present a hypothetical reason. Diffusion is good at modeling the data distribution, where the simple structure of offline RL policy could not perform well. The synthetic data could serve as a medium for distillation of the data modeling capability from diffusion to offline RL. This may take effect in two scenarios. *The first is* for better sample exploitation, which stands for the ability to stitch good trajectories within the dataset. *The second is* for better OOD generalization. The synthetic data generated by diffusion is likely to lie within a range that could be considered a "reasonable guess", allowing the policy to perform robustly when encountered with these unseen cases during evaluation. As a preliminary analysis, we visualize the distribution of single-step reward from the baseline CQL, CQL trimming at 50% percentage, and CQL trained with SER augmentation.

As shown in Figure 7, the performance improvement of SER augmentation primarily stems from the increased proportion of reward values near the maximum reward in the dataset. The increase in rewards that are close to or exceed the maximum value clearly indicates an enhancement in the policy's OOD generalization ability. In contrast, the performance loss of trimming at 50% percentage could mainly be attributed to the inability to perform optimal level behavior of the dataset, indicating that trimming low return trajectories also hurts the ability of CQL to perform optimal level behavior. However, the reason here is intertwined, in that it's still unclear whether it is the ability to survive under hard conditions or the ability to perform optimally under good conditions that influences the

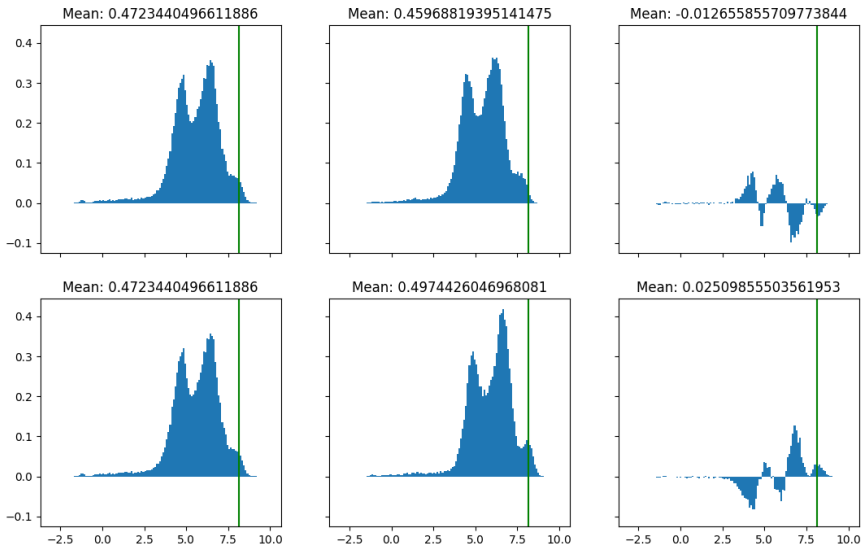



Figure 7: Distribution of single-step reward of CQL. From left to right column: baseline, trimming/augmentation, difference. Top row: trimming at 50% percentage. Bottom row: SER augmentation. The green vertical line denotes the maximum single-step reward in the dataset.

portion of high reward values. It will be interesting to see how exactly diffusion brings performance improvement through more concrete analysis.

## 5.2 EDAC Results

### 5.2.1 Dataset Compression

The practical deployment of offline RL systems often necessitates addressing computational resource constraints. One feasible solution is dataset compression — intelligently trimming the dataset without significantly impacting the performance of baseline algorithms. Our experiment on the EDAC algorithm shows that we can marginally reduce the volume of data without a notable decline in an agent's learning efficacy. By effectively compressing the dataset, one can conserve memory and computational resources, not only facilitating more efficient training cycles but also democratizing the accessibility of RL by reducing the required infrastructure.

### 5.2.2 Diversity and Fidelity

In offline RL, the quantity of data is crucial for enhancing the agent's performance, as more data implies better coverage of the state-action space, which is essential for reducing out-of-distribution error. The diversity and structure of that data play an equally pivotal role in improving the efficacy of training RL agents. If we can take advantage of the function approximation error of diffusion models well, we can use the synthetic data to drive the agent to explore the regions in the sate-action space that is under-visited by the behavior policy. However, over-fitting the synthetic data could mislead the agent, as the approximation error disrupts the real structure of environment transitions in the dataset. We illustrate our finding in Figure 8.

### 5.2.3 Transfer Learning

Transfer learning in the context of offline RL encompasses leveraging knowledge acquired from one domain to enhance performance in another, typically more challenging domain. A straightforward approach might suggest pre-training on synthetic data before fine-tuning with real data. However, our experiment has shown that this simplistic transfer approach often fails. The primary culprit is the

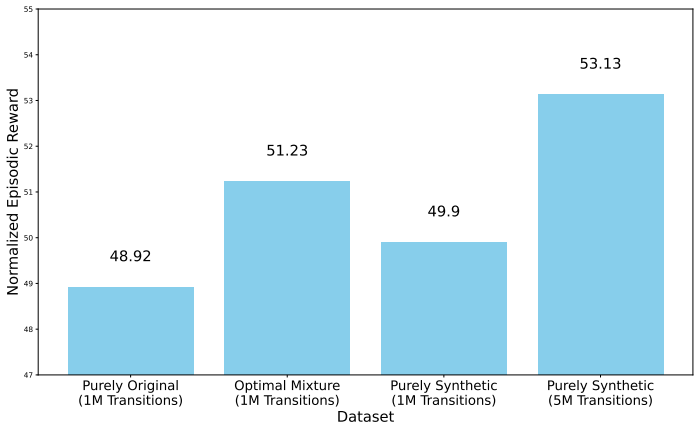



Figure 8: The effect on EDAC performance of the authenticity and size of mixture dataset.

substantial gap between synthetic and real environments brought by imperfect fitting of diffusion models, which can lead to ineffective or even detrimental transfer. Instead, a more nuanced, fine-grained mixture method proves to be more beneficial. This involves integrating synthetic and real data at a more granular level throughout the training process. Such a methodology ensures that the agent incrementally adapts to the real world while still benefiting from the diversity brought by synthetic data. By maintaining a fine-grained balance, the agent progressively refines its policies, thereby achieving superior performance in real-world tasks.

## 6 Conclusion and Future Work

Our study underscores two critical points in offline RL: first, the diversity (quality) of data is as important as its quantity. Diverse data ensures that the learning algorithm encounters a wide range of scenarios, improving its generalization capabilities. Second, the methodology employed in mixing datasets significantly affects the results. Transfer learning processes need to avoid abrupt distribution shifts that could disrupt the training stability.

Our conclusive observations reaffirm the importance of data diversity and appropriate mixture methodologies in offline RL. Ensuring a gradual and balanced introduction of different data types proves essential for robust RL training and effective transfer learning. The balanced approach of interspersing real and synthetic data periodically appears to offer a more stable training pathway compared to a stark transition from all synthetic to all real data.

One possible direction from our current approach is to increase fidelity in synthetic data. We plan to use a forward dynamics model to filter pseudo trajectories, or ensemble diffusion models for importance sampling to address this. Further refinement is needed to ensure the highest data quality.

Another limitation is the dependency on the mixture methodology to balance synthetic and original data, which may require extensive tuning based on the specific offline RL scenario. Future research could explore more robust methods for dynamically adjusting the mixture ratio of synthetic and original data based on real-time performance metrics. A deeper understanding of how the quality of the original dataset influences the choice of mixture ratios and the role of reward noising in offline dataset exploration would be valuable.

Additionally, integrating more sophisticated policy guidance mechanisms, such as those proposed in PGD [4], could help mitigate the distribution shift more effectively.

Moreover, while our approach has shown promising results in offline settings, its application in online RL scenarios remains underexplored. Investigating how diffusion models and offline RL can enhance online RL, as mentioned in SER [8] and Reinforcement Learning with Prior Data (RLPD) [2], could open new avenues for improving algorithm efficiency and effectiveness.

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